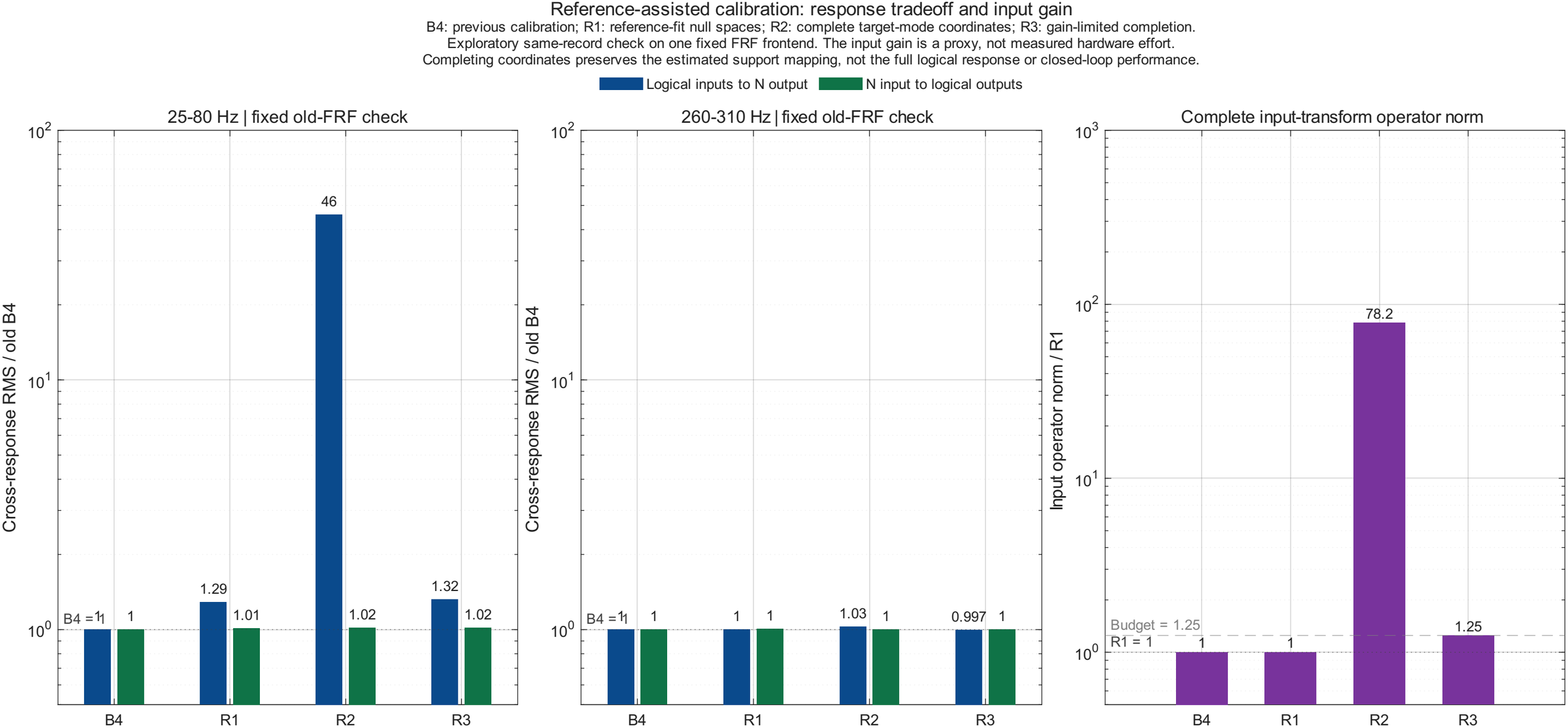

Reference-assisted calibration: response tradeoff and input gain
B4: previous calibration; R1: reference-fit null spaces; R2: complete target-mode coordinates; R3: gain-limited completion.
Exploratory same-record check on one fixed FRF frontend. The input gain is a proxy, not measured hardware effort.
Completing coordinates preserves the estimated support mapping, not the full logical response or closed-loop performance.
Logical inputs to N output
N input to logical outputs
25-80 Hz | fixed old-FRF check
260-310 Hz | fixed old-FRF check
Complete input-transform operator norm
Cross-response RMS / old B4
Input operator norm / R1
B4 = 1
1
1.29
1.01
46
1.02
1.32
1.02
1
1
1
1.03
1
0.997
1
Budget = 1.25
R1 = 1
78.2
1.25
B4
R1
R2
R3